HANCHU ZHOU

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EDUCATION

University of California, Davis, Ph.D. — Sep. 2022 - Present
Electrical and Computer Engineering, Advisor: Junshan Zhang — **Davis, CA**

Zhejiang University, Bachelor — Sep. 2018 - Jun. 2022
Control Science, Advisor: Qinyuan Ren — **Hangzhou, China**

RESEARCH INTERESTS

Robot Learning; World Model; Reinforcement Learning

SELECTED PUBLICATIONS

DuoMind: Enabling Distributed Multi-Robot Coordination with Semantic Communication — *Preprint*
H Zhou, D Gao, H Wang, B Lynch, B Zhao, Q Ma, R Goyal, J Zhang

Agile-WAM: An Agile Tactile World Action Model for Contact-Rich Robot Control — *Preprint*
H Zhou, B Lynch, R Goyal, D Gao, B Kasap, B Zhao, J Zhang

VITA: Vision-to-Action Flow Matching Policy ★ **130+ stars** — *ICLR 2026*
D Gao, B Zhao, A Lee, I Chuang, **H Zhou**, H Wang, Z Zhao, J Zhang, I Soltani

CarDreamer: Open-Source Learning Platform for World-Model-Based Autonomous Driving ★ **360+ stars** — *IEEE IoT Journal*
D Gao*, S Cai*, **H Zhou***, H Wang*, I Soltani, J Zhang

EI-Drive: A Platform for Cooperative Perception with Realistic Communication Models — *IEEE IoT Journal*
H Zhou, E Xie, W Shao, D Gao, M Dong, J Zhang

Personalized RewardBench: Evaluating Reward Models with Human-Aligned Personalization — *COLM 2026*
Q Ma, D Gao, R Cai, B Zhao, **H Zhou**, J Zhang, Z Zhao

IN-RIL: Interleaved Reinforcement and Imitation Learning for Policy Fine-Tuning — *Preprint*
D Gao, H Wang, **H Zhou**, N Ammar, S Mishra, A Moradipari, I Soltani, J Zhang

ON-GOING PROJECTS

Tactile Dexterous Hand Manipulation — Oct. 2026 - Present

· Developing World Action Model for real-world dexterous hands with tactile sensor, with assistance of GPT-6 family models to provide guidance. We aim to avoid the low-efficiency that use LLM for direct control.

Agentic Multi-robot Coordination — Oct. 2026 - Present

· We aim to build an agentic multi-robot system that leverages the embodied reasoning capabilities of LLMs for reliable and efficient task planning and execution.

WORK EXPERIENCE

Tactile World Action Model for Robot — Mar. 2026 - Sep. 2026
Research Intern @ Analog Device — Advisor: Brendan Lynch

SKILLS

Real-World Robots: ALOHA, Franka Research 3, Flevix, Unitree G1, Tesollo DG-5F
Programming: Python, C, C++, MATLAB, HTML, Markdown, C#, JavaScript
Machine Learning: JAX, PyTorch, TensorFlow, Scikit-learn, Pandas, OpenCV, Huggingface
Tools: Open3D, YOLO, Wandb, Matplotlib, Git, ROS, Linux, Conda